TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A PROJECT REPORT

ON

UPTIME TRACKER: WEBSITE MONITORING AND AVAILABILITY TRACKING SYSTEM

Submitted By:

Abhishek Panthee (THA080BCT002)

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Submitted To:

Department of Electronics and Computer Engineering

Thapathali Campus

Kathmandu, Nepal

In partial fulfillment for the award of the

Bachelors Degree in Electronics, Communication and Information Engineering

Under the Supervision of

Er. Rajad Shakya

December 2025

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DECLARATION

We hereby declare that this project entitled **"UPTIME TRACKER: WEBSITE MONITORING AND AVAILABILITY TRACKING SYSTEM"** is our own original work. This project has been prepared and submitted as a DBMS academic project to the Department of Electronics and Computer Engineering, Thapathali Campus. The work presented in this report is genuine and has not been copied from any other source. This work has not been submitted elsewhere for academic evaluation or any other purpose.

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Alin Timelsana	(THA080BCT003)
Kiran Paudel	(THA080BCT019)

Date: December 2025

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ACKNOWLEDGEMENT

We would like to thank Institute of Engineering, Tribhuvan University, for incorporating major project within the curriculum of Bachelor's in Electronics, Communication, and Information Engineering.
Special thanks to our supervisor, **Er. Rajad Shakya**, for his guidance, invaluable feedback, and constant support for the project. His expertise and encouragement have been pivotal in shaping this project. We are also grateful to our project coordinator, **Er. Umesh Kanta Ghimire**, for his continuous support and guidance.
We would like to express our appreciation to the Department of Electronics and Computer Engineering, Thapathali Campus, for their continuous assistance, recommendations during project selection. This has significantly helped in enabling our professional development.
We extend our sincere thanks to all the teachers, friends, and both direct and indirect contributors for their valuable insights, recommendations, and encouragement throughout this journey.

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ABSTRACT

The Uptime Tracker is a web-based monitoring system designed to track website availability and performance metrics in real-time. In today's digital landscape, website downtime can significantly impact productivity, service delivery, and user trust. This project addresses the need for automated monitoring of college websites and web services by implementing a centralized tracking system similar to status.tcioe.edu.np and downdetector.com.

The system features user authentication, website registration and ownership management, automated ping checks at 5-minute intervals, and comprehensive analytics dashboards. Using PostgreSQL database for data storage, the platform monitors HTTP/HTTPS availability, measures response times, and calculates hourly and daily average uptime statistics. The application supports both public and private monitoring modes, allowing website owners to manage visibility of their site's status.

The Uptime Tracker provides real-time status information, historical data visualization, and performance analytics that enable administrators to quickly identify and respond to downtime incidents. This transparency improves user experience and supports data-driven decisions for service reliability improvements.

Keywords—Website Monitoring, Uptime Tracking, Availability Monitoring, Real-time Status, Performance Analytics, PostgreSQL, Web Application, Service Reliability.

TABLE OF CONTENTS

DECLARATION . . . . . **i**

COPYRIGHT . . . . . **ii**

ACKNOWLEDGEMENT . . . . . **iii**

ABSTRACT . . . . . **iv**

LIST OF FIGURES . . . . . **vii**

LIST OF TABLES . . . . . **viii**

LIST OF ABBREVIATIONS . . . . . **ix**

LIST OF SYMBOLS . . . . . **xi**

1. Introduction **1**
- 1.1. Background 1
- 1.2. Problem Statement 1
- 1.3. Objectives 1
- 1.4. Scope and Limitations 2

2. LITERATURE REVIEW **3**
- 2.1. Website Monitoring and Availability 3
- 2.2. Existing Monitoring Solutions 3
- 2.3. Technical Approaches 4
- 2.4. Data Storage 4
- 2.5. Research Gaps 4

3. REQUIREMENTS ANALYSIS **5**
- 3.1. Functional Requirements 5
- 3.2. Non-Functional Requirements 5
- 3.3. System Requirements 5

4. SYSTEM ARCHITECTURE AND METHODOLOGY **6**
- 4.1. System Overview 6
- 4.2. Database Architecture 6

4.3. Monitoring Methodology 7
4.4. User Interaction Flow 7
4.5. Technology Stack 7

5. IMPLEMENTATION DETAILS 8

6. RESULTS AND ANALYSIS 9

7. REMAINING TASKS 10

8. CONCLUSION 11
8.1. Achievements 11
8.2. Limitations 11
8.3. Future Work 11
8.4. Final Remarks 11

APPENDICES 12

A. MATHEMATICAL DERIVATIONS 12
A.1. Proof of Theorem 1: 12
A.2. Proof of Theorem 2: 12
A.3. Gaussian Distribution 12

B. PROJECT BUDGET 13
B.1. Development Equipment 13
B.2. Bill of Materials (BOM) 13
B.3. Software and Cloud Services 14
B.4. Miscellaneous Expenses 14

C. PROJECT TIMELINE 16
C.1. Gantt Chart 16

D. PLAGARISM SUMMARY 18

E. COMPREHENSIVE LaTeX GUIDE 19

REFERENCES 37

LIST OF FIGURES

C.1. Project Gantt Chart 17
E.1. System architecture showing main components and their interactions. 23
E.2. Model performance during training showing both training and validation metrics. 25

LIST OF TABLES

2.1. Comparison of Website Monitoring Solutions 3
B.1. Development Equipment Cost 13
B.2. Bill of Materials 14
B.3. Software and Cloud Services Cost 14
B.4. Estimated Cost Breakdown 15
E.1. Experimental parameters and their values. 25

LIST OF ALGORITHMS

1. Genetic Algorithm for Optimization 27

LIST OF ABBREVIATIONS

IEEE	Institute of Electrical and Electronics Engineers

1. INTRODUCTION

1.1. Background

In today's digital era, website availability and uptime are critical for businesses, educational institutions, and organizations. Websites serve as the primary interface for information dissemination, service delivery, and communication. Any downtime can result in loss of productivity, revenue, and user trust.

Educational institutions like Thapathali Campus rely heavily on multiple web services for student information systems, result portals, library services, and administrative functions. However, monitoring the availability of these services manually is time-consuming and inefficient. There is a need for automated monitoring solutions that can track website uptime, detect downtime events, and provide real-time status information to users.

Similar to platforms like status.tcioe.edu.np and downdetector.com, an uptime tracking system can provide valuable insights into website availability, response times, and performance metrics over time.

1.2. Problem Statement

Currently, there is no centralized system to monitor and track the uptime status of college websites and services. When a website goes down, users are often unaware whether the issue is temporary or long-term. System administrators lack historical data on uptime patterns, average response times, and downtime incidents.

The absence of such a monitoring system leads to:

- Delayed response to downtime incidents
- Lack of transparency about service availability
- No historical data for performance analysis
- Difficulty in identifying patterns of downtime
- Poor user experience due to unavailability information

1.3. Objectives

The primary objectives of this project are:

1. **Develop an automated website monitoring system** that can track the uptime status of multiple websites in real-time.

2. **Implement periodic ping checks** to monitor website availability at regular intervals (every 5 minutes).
3. **Store and analyze historical data** including response times, average hourly uptime, and daily availability metrics.
4. **Create a user-friendly dashboard** to display current website status and historical analytics.
5. **Implement user authentication** to allow website owners to register and monitor their sites.
6. **Provide public and private monitoring options** for different use cases.

1.4. Scope and Limitations

1.4.1. Scope

The Uptime Tracker system will include:

- User registration and authentication system
- Website registration and ownership management
- Automated ping monitoring at 5-minute intervals
- Response time measurement and logging
- Hourly and daily average calculation
- Public status page for monitored websites
- Analytics dashboard showing uptime statistics
- Historical data visualization
- PostgreSQL database for data storage

1.4.2. Limitations

- The system monitors only HTTP/HTTPS website availability and does not check specific application functionality
- Monitoring frequency is limited to 5-minute intervals to balance server load and data accuracy
- The system does not provide detailed performance metrics like page load times or resource usage
- No automated alerting or notification system for downtime events in the initial version
- Geographic monitoring is limited to a single server location

2. LITERATURE REVIEW

2.1. Website Monitoring and Availability

Website availability monitoring is critical for modern web infrastructure, with IT downtime costing approximately $5,600 per minute [1]. Monitoring systems employ active probing—sending periodic HTTP requests to verify availability by checking response codes and response times [2].

Patterson et al. [3] defined uptime as the percentage of time a system remains operational, typically expressed as "nines" (e.g., 99.9% equals 8.76 hours downtime per year) in Service Level Agreements.

2.2. Existing Monitoring Solutions

2.2.1. Commercial Platforms

Pingdom offers comprehensive monitoring with global probe locations [4]. UptimeRobot provides 5-minute monitoring intervals with a freemium model [5]. Downdetector aggregates user reports rather than active monitoring [6].

2.2.2. Open-Source Solutions

Uptime Kuma is a popular self-hosted solution with Docker deployment [7]. Nagios and Zabbix provide enterprise-grade capabilities but require significant expertise [8, 9]. Prometheus with Blackbox Exporter enables HTTP probing within cloud-native environments [10].

Table 2.1 compares key features of existing solutions.

Table 2.1. *Comparison of Website Monitoring Solutions*

Solution	Type	Interval	Status Page	Self-Hosted
Pingdom	Commercial	1 min	Yes	No
UptimeRobot	Commercial	5 min	Yes	No
Uptime Kuma	Open-Source	1 min	Yes	Yes
Nagios	Open-Source	1 min	Limited	Yes
Our System	Open-Source	5 min	Yes	Yes

2.3. Technical Approaches

HTTP monitoring sends requests to URLs and analyzes response status codes, response time, and SSL certificate validity [2]. ICMP ping provides basic reachability but doesn't verify application functionality [11]. Five-minute intervals balance detection speed with resource consumption [12].

2.4. Data Storage

PostgreSQL handles time-series monitoring data efficiently with temporal queries and aggregations [13]. For moderate data volumes, it provides adequate performance with simpler deployment than specialized databases like InfluxDB [14].

2.5. Research Gaps

Existing solutions have limitations for educational institutions:

- Commercial solutions impose recurring subscription costs
- Enterprise tools (Nagios, Zabbix) require significant expertise
- Cloud-based monitoring cannot detect internal network issues
- Limited customization for institutional requirements

Our Uptime Tracker addresses these gaps with a simple, self-hosted solution for institutions like Thapathali Campus.

3. REQUIREMENTS ANALYSIS

3.1. Functional Requirements

- Users can create accounts and log in securely.
- Users can add their websites to be monitored.
- The system checks each website every 5 minutes automatically.
- Each check records whether the site is up and how fast it responds.
- The system calculates average uptime per hour and per day.
- Anyone can view public status pages showing website health.
- Users see a dashboard with charts and uptime history.
- Users can choose to make their monitors public or private.

3.2. Non-Functional Requirements

- **Speed:** Pages and API calls load in under 500 milliseconds.
- **Reliability:** The monitoring service runs 99% of the time.
- **Capacity:** The system can handle 100+ websites being monitored.
- **Mobile-friendly:** The dashboard works well on phones and tablets.
- **Security:** User passwords are encrypted before storing.

3.3. System Requirements

3.3.1. Hardware Requirements

- **Minimum:** Any 64-bit computer with 4 GB RAM and 10 GB storage
- **Recommended:** Dual-core processor, 8 GB RAM, SSD for faster performance

3.3.2. Software Requirements

- Operating System: Windows, Linux, or macOS
- Backend: Bun.js (JavaScript runtime)
- Frontend: Next.js with React
- Database: PostgreSQL

4. SYSTEM ARCHITECTURE AND METHODOLOGY

4.1. System Overview

The Uptime Tracker system consists of three main components: a web application frontend for user interaction, a backend server for business logic and API handling, and a PostgreSQL database for persistent storage. The system also includes an automated monitoring service that periodically checks website availability.

4.2. Database Architecture

The system uses PostgreSQL database with the following schema:

4.2.1. Users Table

Stores user account information including name, email, and authentication credentials. Each user is identified by a unique ID and email address.

4.2.2. Ownership Table

Maintains relationships between websites and their owners. Includes a boolean flag for public/private visibility control. This table links website URLs to user accounts.

4.2.3. Analytics Table

Records individual ping results for monitored websites. Each entry includes:

- Website URL (foreign key to ownership)
- Response time (ping5) in milliseconds
- Timestamp of the check

4.2.4. Average Hour Table

Stores aggregated hourly statistics including average response time for each website within each hour period.

4.2.5. Average Day Table

Maintains daily average statistics, providing long-term trends and historical performance data.

4.3. Monitoring Methodology

4.3.1. Ping Check Process

The monitoring service performs HTTP/HTTPS requests to registered websites every 5 minutes. For each check:

1. Send HTTP request to the website URL
2. Measure response time
3. Record success/failure status
4. Store result in analytics table
5. Update hourly and daily averages

4.3.2. Data Aggregation

Hourly and daily averages are calculated using SQL aggregate functions on the analytics data. This provides efficient querying for dashboard displays and trend analysis.

4.4. User Interaction Flow

1. User registers and creates an account
2. User adds website URLs to monitor
3. System begins automated monitoring
4. User views real-time status on dashboard
5. User accesses historical analytics and reports

4.5. Technology Stack

- **Database:** PostgreSQL 18.1
- **Backend:** (To be determined - Python/Node.js)
- **Frontend:** (To be determined - React/Vue.js)
- **Monitoring:** Scheduled cron jobs or background workers

5. IMPLEMENTATION DETAILS

Note: This begining of this chapter should be mid-point of your entire rport content. The implementation detials and result part should approximately be 50-60% of total content. Describe how design was translated to working system with sufficient detail for replication. Some relevant sections might be given below. You can heavilty customize based on your project type.

6. RESULTS AND ANALYSIS

The Results and Analysis chapter presents the outcomes of your project and critically analyzes them. This chapter demonstrates whether your system meets requirements, compares performance against objectives and baselines, and interprets findings. Results should be presented objectively with appropriate visualizations, while analysis provides context and explanation. This chapter should be full of charts, tables, and figures illustrating your system's performance and behaviour.

- Always reference figures and tables in the text before they appear.
- Use the cleveref package; write `\cref{fig:label}` or `\Cref{tab:label}` for automatic formatting (add `\usepackage{cleveref}` to the preamble).
- Keep captions descriptive and use unique labels per chapter (e.g., `fig:my-figure`, `tab:results`).

7. REMAINING TASKS

CHAPTER GUIDE: Purpose: describe unfinished work, risks, and a realistic plan. Must include: What is completed vs pending, Remaining milestones, Risks/blockers + mitigation. LaTeX: Use itemize for checklists; enumerate for ordered milestones.

8. CONCLUSION

The Conclusion chapter summarizes your project, highlights key achievements, discusses limitations, and suggests future work. It should provide closure to your report and leave readers with a clear understanding of what you accomplished and its significance.

8.1. Achievements

8.2. Limitations

8.3. Future Work

8.4. Final Remarks

Highlight major accomplishments: Briefly recap the problem you addressed, your approach, and what you built. This should be 1-2 paragraphs summarizing the entire project without going into technical details.

- Whether objectives were met - Performance metrics achieved - Novel contributions or innovations - Successful integration or deployment Be specific and quantitative where possible.

Acknowledge limitations of your work honestly: - Technical constraints encountered - Performance limitations - Scope restrictions - Areas where further improvement is needed

FUTURE WORK: Suggest concrete directions for extending your project: - Additional features that could be implemented - Performance improvements that could be made - Alternative approaches that could be explored - Scaling or deployment in real-world settings - Research questions that emerged from your work

End with a brief statement about the significance of your work and its potential impact.

LaTeX quick refs: - Keep conclusion concise; avoid introducing new results. - If referencing key results: point to earlier figures/tables via

A. MATHEMATICAL DERIVATIONS

You can include the Derivation and mathematical part of your project that is not necessary to be included in the main report here. To provide reader with the complete mathematical background of your project, you can include the derivations, proofs, and detailed explanations of theorems or formulas used in your project. and then you can cross reference them in the main report.

A.1. Proof of Theorem 1:

Convex Optimization Problem can be solved using Gradient Descent Method.

A.2. Proof of Theorem 2:

Convergence of Gradient Descent Method for Convex Functions.

A.3. Gaussian Distribution

The Gaussian distribution, also known as the normal distribution, is a continuous probability distribution characterized by its bell-shaped curve. It is defined by its mean (μ) and standard deviation (σ). The probability density function (PDF) of a Gaussian distribution is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

B. PROJECT BUDGET

The total budget required for the successful completion of the project is estimated to be between $330 and $390. This budget covers various aspects of the project, including hardware purchases, component materials, software services, and miscellaneous expenses.

B.1. Development Equipment

This section lists essential hardware and tools that must be purchased for development and testing. These are items not available in the lab or not part of the final product, but necessary for the project.

Table B.1. *Development Equipment Cost*

Item	Specification	Qty	Unit Cost	Total
Development Board	Raspberry Pi 4 (4GB)	1	$55	$55
Testing Equipment	Digital Multimeter	1	$25	$25
Sensors	Temperature/Humidity Sensor	2	$10	$20
			Subtotal	**$100**

Note: Replace the example entries above with actual equipment needed for your project.

B.2. Bill of Materials (BOM)

Keep this only if relevant to your project. This section details all components that will be integrated into the final product/system. These are the materials that constitute the deliverable hardware (if any).

Table B.2. *Bill of Materials*

Component	Specification	Qty	Unit Cost	Total
Microcontroller	ESP32 DevKit	1	$10	$10
Sensor	DHT22 Sensor	2	$5	$10
Actuator	Servo Motor SG90	2	$3	$6
Power Supply	5V/2A Adapter	1	$8	$8
PCB	Custom PCB (100x100mm)	1	$15	$15
Enclosure	ABS Plastic Box	1	$12	$12
			Subtotal	**$61**

Note: Replace the example entries above with actual components for your project.

B.3. Software and Cloud Services

This section covers all software tools, cloud computing resources, and subscription services required for the project.

Table B.3. *Software and Cloud Services Cost*

Service	Description	Cost
Open-Source Software	Python, NumPy, SciPy, TensorFlow, PyTorch, etc.	Free
Cloud Computing	Google Colab Pro (GPU access for training)	$10/month
Cloud Storage	Google Drive (100GB)	$2/month
API Services	Third-party API usage (if applicable)	$15
IEEE Access	Student subscription for research papers	$15
	Total (3 months)	**$66**

Note: Adjust the entries above based on your actual software and service requirements.

B.4. Miscellaneous Expenses

This section includes other project-related costs such as documentation, printing, and administrative expenses.

The complete budget estimation is shown in Table B.4. The budget is designed to ensure that all required resources are available for the successful completion of the project.

This expense breakdown in Table B.4 incorporates the first half of the project.

Note: All tables in this document use the `booktabs` package for professional table formatting, as per IEEE style guidelines. Use `\toprule`, `\midrule`, and `\bottomrule` instead of `\hline` for better visual appearance.

Table B.4. *Estimated Cost Breakdown*

Category	Item/Description	Estimated Cost
Software	Python, NumPy, SciPy, DEAP, SQLite, Matplotlib, Seaborn, Gym (for RL), and other required libraries (all open-source)	Free
Cloud Computing	**Google Cloud Platform (GCP)**: - NVIDIA T4: $0.35/hour - NVIDIA V100: $2.48/hour - NVIDIA A100 (40GB): $4.27/hour **Google Colab Pro**: - $9.99/month for 100 compute units - Approx. 13 units/hour for A100 usage	Estimated $300-$350
IEEE Access	Annual subscription for accessing IEEE research articles and journals	$15 (student subscription)
Miscellaneous	Printing, documentation, report binding	$15-$25
Total Estimated Cost		**$330-$390**

C. PROJECT TIMELINE

The project will be executed over a period of approximately 6 months, divided into several key phases. Each phase includes specific tasks and milestones to ensure steady progress toward project completion.

C.1. Gantt Chart

gantt should clearly show the year month, week, and tasks with their durations and dependencies, with milestones also properly marked. This is the estimated timeline for the project, detailing the key phases and milestones. The project is structured into several stages, each with specific tasks and deliverables.

Gantt Chart Placeholder here:

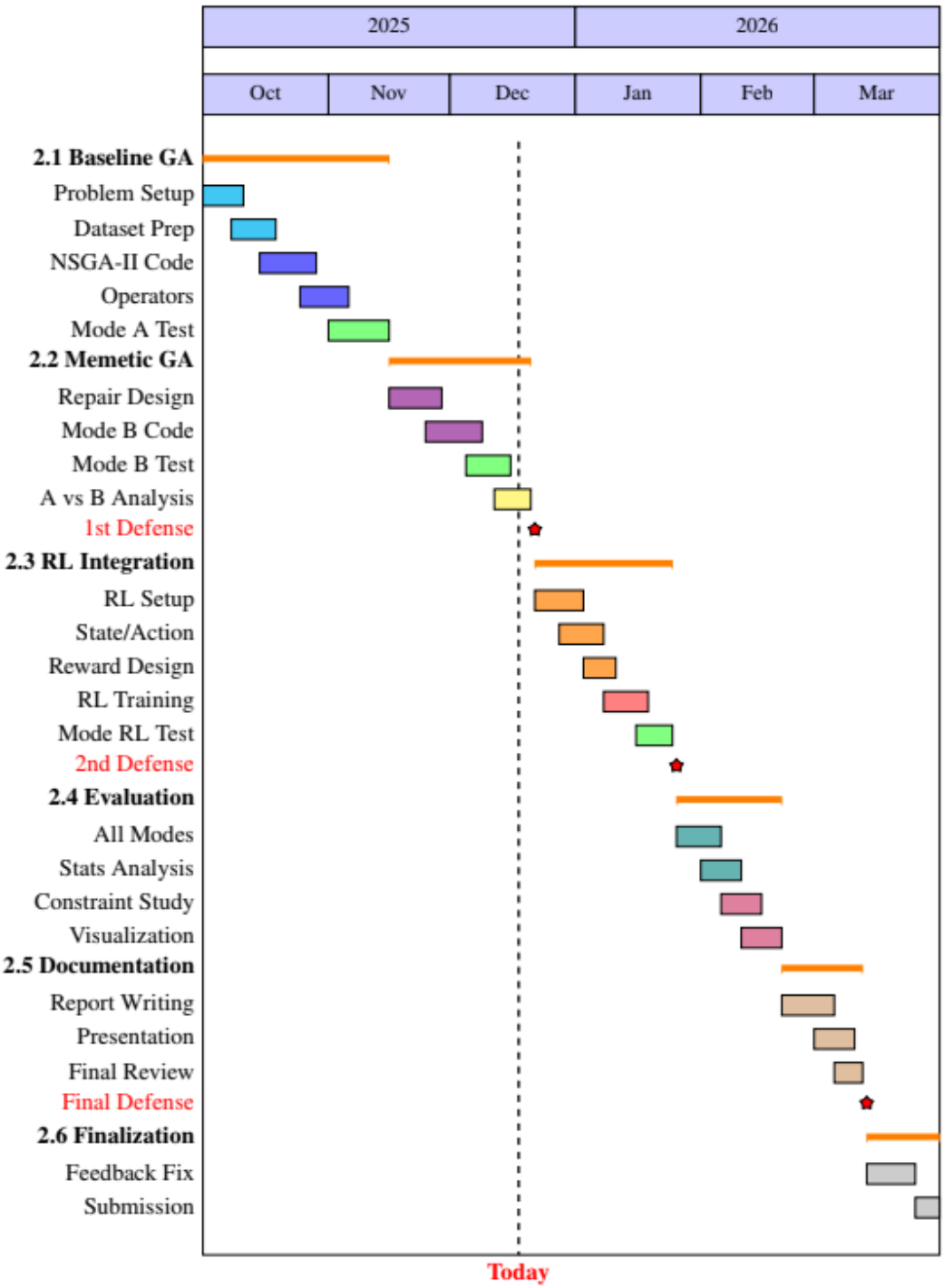

Figure 8-1 Project Timeline

Figure C.1. *Project Gantt Chart*

D. PLAGARISM SUMMARY

Keep all the pages of the plagarism summary document provided by the university here. It can run for multiple pages.

E. COMPREHENSIVE LaTeX GUIDE

This chapter provides a comprehensive guide to using this LaTeXtemplate effectively. It covers essential LaTeXcommands, formatting guidelines, and practical examples with rendered output to help you write high-quality technical documents following Institute of Electrical and Electronics Engineers (IEEE) standards.

Template Philosophy

This template is designed with three core principles:

- **IEEE Compliance**: All formatting adheres to IEEE Transactions standards for academic publications
- **Automation**: Cross-references, numbering, and bibliography are handled automatically
- **Separation of Concerns**: Content is separated from formatting—you focus on writing, the template handles presentation

The template leverages modern LaTeXpackages (`newtxtext`, `newtxmath`, `cleveref`, `glossaries`, `booktabs`, `algorithmicx`) to produce professional, publication-ready documents.

Quick Start Guide

Initial Setup

1. **Install LaTeXDistribution**:
   - Windows: MiKTeX or TeX Live
   - macOS: MacTeX
   - Linux: TeX Live (via package manager)
2. **Configure Your Information**: Edit `vars.tex` with your personal details, project title, supervisors, and institutional information.
3. **Compilation Sequence**: To properly generate all references, bibliography, and glossaries, compile in this order:

```
pdflatex main.tex
bibtex main
makeglossaries main
```

```
pdflatex main.tex
pdflatex main.tex
```

Most LaTeXeditors (TeXstudio, Overleaf, VS Code with LaTeXWorkshop) handle this automatically. Using glossaries package on VS Code requires setting up shell escape.

4. **Project Structure**:

```
main.tex              - Main document file
vars.tex              - Your personal information
thapathaliece.cls     - Template class (do not modify)
references.bib        - Bibliography database
src/
  frontmatter/        - Abstract, acknowledgments, etc.
  chapters/           - Your chapter content files
  backmatter/         - Appendices, supplementary material
  images/             - All figures and diagrams
```

Essential LaTeXConcepts

Document Structure

Each chapter file should follow this structure:

```
\chapter{CHAPTER TITLE}

Introduction paragraph for the chapter...

\section{Section Title}

Content for this section...

\subsection{Subsection Title}

Detailed content...

\subsubsection{Subsubsection Title}

Fine-grained details...
```

Text Formatting

Basic text formatting commands:

```
\textbf{bold text}
\textit{italic text}
\texttt{monospace/code text}
\underline{underlined text}
\emph{emphasized text}
```

Rendered examples:

- **bold text** for emphasis
- *italic text* for terms and titles
- `monospace text` for code, filenames, and commands
- <u>underlined text</u> (use sparingly)
- *emphasized text* (context-aware emphasis)

Lists

Three types of lists are available:

```
\begin{itemize}
    \item First bullet point
    \item Second bullet point
    \item Third bullet point
\end{itemize}

\begin{enumerate}
    \item First numbered item
    \item Second numbered item
    \item Third numbered item
\end{enumerate}

\begin{description}
    \item[Term 1] Definition of term 1
    \item[Term 2] Definition of term 2
\end{description}
```

Mathematical Notation

Inline vs Display Math

LaTeXprovides two modes for mathematics:

```
Inline math: $E = mc^2$ appears in text flow.

Display math:
\[
E = mc^2
\]
appears centered on its own line.
```

Rendered: Inline math like $E = mc^2$ flows with text, while display math is centered:

$$E = mc^2$$

Numbered Equations

For important equations that need referencing:

```
\begin{equation}
    f(x) = \sum_{i=1}^{n} w_i \cdot x_i + b
    \label{eq:linear_model}
\end{equation}

Reference using: \cref{eq:linear_model}
```

Rendered:

$$f(x) = \sum_{i=1}^{n} w_i \cdot x_i + b \tag{E.1}$$

where $f(x)$ is the output, w_i are weights, x_i are inputs, and b is the bias term. Reference as: Eq. (E.1).

IEEE Mathematical Notation

Follow IEEE conventions for mathematical symbols:

```
Variables (italic):     $x$, $y$, $n$
Vectors (bold):         $\vect{v}$, $\vect{x}$
Matrices (bold):        $\matr{A}$, $\matr{X}$
Sets (calligraphic):    $\set{S}$, $\set{P}$
Functions (roman):      $\sin(x)$, $\log(n)$
```

Rendered: Variables x, y, n; vectors $\mathbf{v}$, $\mathbf{x}$; matrices $\mathbf{A}$, $\mathbf{X}$; sets $\mathcal{S}$, $\mathcal{P}$; functions $\sin(x)$, $\log(n)$.

Figures and Images

Including Figures

Basic figure syntax following IEEE standards:

```
\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.8\textwidth]{src/images/diagram.png}
    \caption{System architecture showing main components.}
    \label{fig:architecture}
\end{figure}

Reference using: \cref{fig:architecture}
```

Rendered output:

Figure E.1. *System architecture showing main components and their interactions.*

As shown in Fig. E.1, the architecture demonstrates proper figure formatting with caption below.

Key points:

- `[htbp]` specifies placement: here, top, bottom, page
- `\centering` centers the figure
- Width can be: `0.5\textwidth`, `10cm`, `width=\textwidth`
- Caption appears **below** figure (IEEE standard)
- Always use `\label` for cross-referencing
- Use `\cref` for automatic "Fig." prefix

Subfigures

For multiple related figures:

```
\begin{figure}[htbp]
    \centering
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image1.png}
        \caption{First subfigure}
        \label{fig:sub1}
    \end{subfigure}
    \hfill
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image2.png}
        \caption{Second subfigure}
        \label{fig:sub2}
    \end{subfigure}
    \caption{Overall caption for both subfigures.}
    \label{fig:combined}
\end{figure}
```

Reference individual subfigures: `\cref{fig:sub1}` or the entire figure: `\cref{fig:combined}`.

Rendered output:

The subfigures (Figs. E.2a and E.2b) show different aspects of the same experiment, while Fig. E.2 references the entire figure.

Tables

IEEE-Compliant Tables

Critical: Use `booktabs` package rules only. toprule, midrule and bottom rule. Never use vertical lines (for column separation) or `\hline`. Look at this example! The top line and

(a) *Training accuracy over epochs*

(b) *Validation accuracy over epochs*

Figure E.2. *Model performance during training showing both training and validation metrics.*

bottom lines are thick while the middle line is thin. This consistency is required throughout the report. Always label and cross reference all tables in the document.

Table E.1. *Experimental parameters and their values.*

Parameter	**Value**	**Unit**
Population Size	100	individuals
Generations	500	iterations
Mutation Rate	0.05	probability

```
\begin{table}[htbp]
    \centering
    \caption{Experimental parameters and their values.}
    \label{tab:parameters}
    \begin{tabular}{lcc}
        \toprule
        \textbf{Parameter} & \textbf{Value} & \textbf{Unit} \\
        \midrule
        Population Size    & 100            & individuals   \\
        Generations        & 500            & iterations    \\
        Mutation Rate      & 0.05           & probability   \\
        \bottomrule
    \end{tabular}
\end{table}
```

```
Reference using: \cref{tab:parameters}
```

Key points:

- Caption appears **above** table (IEEE standard)
- Use `\toprule`, `\midrule`, `\bottomrule` only
- Column alignment: `l` (left), `c` (center), `r` (right)
- Bold headers: `\textbf{Header}`
- Never use `|` for vertical lines

Code and Pseudocode in Academic Reports

IEEE Standard for Formal Reports

Important Note: For formal academic reports following IEEE standards, pseudocode using the `algorithmicx` package is **strongly preferred** over direct code embedding.

Why Pseudocode is Preferred

- **Language-agnostic**: Focuses on logic rather than implementation details
- **Professional presentation**: Consistent with academic publication standards
- **Better readability**: Semantic structure with clear control flow
- **Space efficient**: Avoids lengthy implementation details
- **IEEE compliant**: Follows academic formatting conventions

When Direct Code May Be Acceptable

Direct code listings are generally **discouraged** in formal academic reports. However, they may be appropriate in:

- **Appendices**: For reference or reproducibility
- **Technical documentation**: Implementation guides or manuals
- **Software-focused papers**: When code is the primary contribution
- **API demonstrations**: Showing specific library usage

Inline Code References

For mentioning function names, variables, or commands in text:

```
Use the \texttt{calculate\_fitness()} function to evaluate solutions.
Set the \texttt{population\_size} parameter to 100.
```

Rendered: Use the `calculate_fitness()` function to evaluate solutions. Set the `population_size` parameter to 100.

Algorithms and Pseudocode

Algorithm Environment

For formal algorithms, use `algorithmicx` with `algpseudocode`:

```
\begin{algorithm}[htbp]
\caption{Genetic Algorithm for Optimization}
\label{alg:genetic}
\begin{algorithmic}[1]
\Require Population size $N$, generations $G$
\Ensure Optimized solution $\mathbf{x}^*$
\State Initialize population $P \gets \text{random}(N)$
\For{$g = 1$ to $G$}
    \State Evaluate fitness for all individuals in $P$
    \State $P' \gets \text{selection}(P)$
    \State $P'' \gets \text{crossover}(P')$
    \State $P \gets \text{mutation}(P'')$
\EndFor
\State \Return best individual from $P$
\end{algorithmic}
\end{algorithm}

Reference using: \cref{alg:genetic}
```

Rendered output:

Algorithm 1 Genetic Algorithm for Optimization

Require: Population size N, generations G
Ensure: Optimized solution $\mathbf{x}^*$

```
Initialize population P ← random(N)
for g = 1 to G do
    Evaluate fitness for all individuals in P
    P' ← selection(P)
    P'' ← crossover(P')
    P ← mutation(P'')
end for
return best individual from P
```

Key commands:

- `\State`: Single statement
- `\If....\EndIf`: Conditional blocks
- `\For....\EndFor`: Loops
- `\While....\EndWhile`: While loops
- `\Require`: Input requirements
- `\Ensure`: Output guarantee
- `\Return`: Return statement

Citations and References

Adding References

Add entries to `references.bib` in BibTeX format:

```
@article{smith2020machine,
    author  = {Smith, John and Doe, Jane},
    title   = {Machine Learning for Optimization},
    journal = {IEEE Transactions on Neural Networks},
    year    = {2020},
    volume  = {31},
    number  = {5},
    pages   = {1234--1245}
}
```

Citing References

Use `\cite` command:

```
Machine learning has shown promising results \cite{smith2020machine}.
Multiple citations can be combined \cite{ref1,ref2,ref3}.
```

Rendered: Citations appear as numbers in square brackets: [1], [2–5].

Acronyms and Glossaries

Defining Acronyms

Add definitions to `src/frontmatter/abbreviations.tex`:

```
\newacronym{ml}{ML}{Machine Learning}
\newacronym{ai}{AI}{Artificial Intelligence}
\newacronym{nn}{NN}{Neural Network}
```

Using Acronyms

```
\gls{ml}      - First use: Machine Learning (ML)
              - Later uses: ML
\glspl{nn}    - Plural form: NNs
\Gls{ai}      - Capitalized: Artificial Intelligence (AI)
\acrshort{ml} - Always shows: ML
\acrlong{ml}  - Always shows: Machine Learning
```

Benefits:

- Automatic expansion on first use
- Consistent abbreviation usage throughout document
- Automatic generation of abbreviations list
- Prevents undefined acronym errors

Cross-Referencing

Modern Cross-References with Cleveref

This template uses `cleveref` package for intelligent cross-referencing:

```
\cref{fig:arch}        → Fig. 1.1
\cref{tab:results}     → Table 2.3
\cref{eq:fitness}      → Eq. (3.4)
\cref{alg:nsga}        → Algorithm 4.1
\cref{sec:method}      → Section 5.2
\cref{chap:intro}      → Chapter 1

Multiple references:
\cref{fig:a,fig:b,fig:c}    → Fig. 1.1, 1.2, and 1.3
\cref{tab:x,eq:y}           → Table 2.1 and Eq. (3.5)
```

Advantages over manual referencing:

- Automatic prefix (Fig., Table, Eq., etc.)
- Automatic number updating when content changes
- Intelligent grouping of multiple references
- Hyperlinks to referenced elements (in PDF)

Special Characters and Typography

Dashes

LaTeXdistinguishes between three types of dashes:

```
Hyphen:     state-of-the-art (single -)
En-dash:    pages 10--20 (double --)
Em-dash:    results---as shown---improved (triple ---)
```

Rendered: state-of-the-art, pages 10–20, results—as shown—improved.

Quotation Marks

Use proper typographic quotes:

```
``double quotes''
`single quotes'
```

Rendered: "double quotes" and 'single quotes'.
Never use straight quotes copied from Word or text editors.

Special Symbols

Common special characters require backslashes:

```
\%    - Percent sign
\$    - Dollar sign
\&    - Ampersand
\_    - Underscore
\#    - Hash/pound
\{    - Left brace
\}    - Right brace
\~{}  - Tilde
```

Best Practices and Workflow

File Organization

- **One chapter per file**: Keep chapters in separate files for easier management
- **Descriptive filenames**: Use names like `methodology.tex`, `results.tex`
- **Image organization**: Store images in `src/images/` with subdirectories if needed
- **Version control**: Use Git to track changes and collaborate

Compilation Tips

- **Clean auxiliary files**: Regularly delete `.aux`, `.log`, `.toc` files if errors persist
- **Full recompile**: Run complete sequence (pdflatex → bibtex → makeglossaries → pdflatex × 2)
- **Check errors**: Read error messages carefully—they indicate file and line number
- **Incremental compilation**: Compile frequently to catch errors early

Common Mistakes to Avoid

- **UTF-8 characters**: Avoid smart quotes, em-dashes, and special characters copied from Word
- **Unclosed braces**: Every `{` must have matching `}`
- **Missing labels**: Always add `\label` to figures, tables, equations, and algorithms
- **Unreferenced elements**: Every figure, table, and equation must be referenced in text
- **Manual numbering**: Never type "Fig. 1" manually—use `\cref`
- **Vertical lines in tables**: Use `booktabs` rules, not `|` or `\hline`

Quality Checklist

Before final submission, verify:

- ☐ Document compiles without errors or warnings
- ☐ All figures, tables, and equations are referenced in text
- ☐ All citations appear in bibliography
- ☐ Abbreviations are defined and used consistently
- ☐ Cross-references show numbers (no "??")
- ☐ Table of contents has correct page numbers
- ☐ All code listings have captions and proper formatting
- ☐ Mathematical notation follows IEEE conventions
- ☐ No UTF-8 encoding errors in bibliography
- ☐ PDF bookmarks and hyperlinks work correctly

Advanced Features

Custom Commands

The template provides custom commands for convenience:

```
\vect{v}      - Bold vector notation
\matr{A}      - Bold matrix notation
```

```
\set{S}       - Calligraphic set notation
\ie           - i.e.,
\eg           - e.g.,
\etc          - etc.
```

Units with siunitx

For proper unit formatting:

```
\SI{5}{\meter}
\SI{100}{\kilo\gram}
\SI{3.5}{\giga\hertz}
\SIrange{10}{20}{\celsius}
```

Rendered: 5 m, 100 kg, 3.5 GHz, 10–20°C.

Getting Help

Resources

- **This document**: Read all guideline chapters thoroughly
- **LaTeX Stack Exchange**: `tex.stackexchange.com` for specific questions
- **Overleaf Documentation**: Comprehensive LaTeXtutorials
- **Template class file**: See `thapathaliece.cls` for implementation details

Common Issues

Bibliography not appearing Run: pdflatex → bibtex → pdflatex × 2

Acronyms not expanding Run: pdflatex → makeglossaries → pdflatex × 2

Undefined references (??) Compile multiple times until stable

Missing images Check file paths and extensions (.png, .png, .jpg)

UTF-8 errors Replace special characters in .bib file with LaTeXequivalents

WRITING GUIDELINES

This section provides comprehensive guidance on technical writing following IEEE standards and best practices for academic publications [15, 16].

Avoiding Common Mistakes

Grammar and Style

- **Avoid**: "The data was analyzed" → **Use**: "The data were analyzed" (data is plural)

- **Avoid**: "very unique", "more optimal" → **Use**: "unique", "optimal" (absolute adjectives)
- **Avoid**: "In this paper" → **Use**: "In this report" or "In this work"
- **Avoid**: Contractions (don't, can't) → **Use**: Full forms (do not, cannot)

Technical Writing

- Define terms before using them extensively
- Maintain consistency in terminology throughout
- Use specific numbers rather than vague terms: "15% improvement" not "significant improvement"
- Explain acronyms at first use, even if well-known in the field
- Cross-reference related sections appropriately using `\cref{}` commands

COMPILING AND TROUBLESHOOTING

This section provides practical guidance on compiling your report and resolving common issues.

Compilation Process

Standard Compilation Sequence

To properly compile this report with all references, use this sequence:

1. **pdfLaTeX**: First pass to generate auxiliary files
2. **bibtex**: Process bibliography references
3. **makeglossaries**: Generate abbreviations list
4. **pdfLaTeX**: Second pass to include bibliography
5. **pdfLaTeX**: Third pass to resolve all cross-references

Command Line Compilation

Open terminal in report directory and run:

```
pdflatex main.tex
bibtex main
makeglossaries main
pdflatex main.tex
pdflatex main.tex
```

Using LaTeX Editors

Most LaTeX editors have "Build" buttons that handle the sequence automatically:

- **TeXstudio**: Press F5 or click "Build & View"
- **Overleaf**: Compiles automatically on save
- **VS Code**: With LaTeX Workshop extension, use "Build LaTeX project"

Common Compilation Errors

Missing Package Errors

Error: `! LaTeX Error: File 'package.sty' not found`
Solution: Install the missing package:

- TeX Live: `tlmgr install package`
- MiKTeX: Will prompt to install automatically
- Or install full LaTeX distribution

Undefined Reference Warnings

Warning: `LaTeX Warning: Reference 'fig:example' undefined`
Solution: Run pdfLaTeX multiple times (2–3) after adding new labels. References require multiple passes to resolve.

Citation Undefined

Warning: `Citation 'smith2020' undefined`
Solution:

- Check that citation key exists in `references.bib`
- Run bibtex step
- Run pdflatex again

Glossary Not Appearing

Problem: Abbreviations list is empty or doesn't appear
Solution:

- Ensure you've used `\gls{}` commands in text
- Run `makeglossaries` command
- Run pdfLaTeX again
- Glossaries only shows acronyms actually used in document

Font Errors

Error: Font shape warnings or missing font files
Solution:

- Ensure newtx package is installed: `tlmgr install newtx`
- Update LaTeXdistribution to latest version
- Run `updmap` or `updmap-sys` to refresh font database

File Organization Issues

File Not Found Errors

Error: `! LaTeXError: File 'src/chapters/intro.tex' not found`
Solution:

- Verify file exists in correct directory
- Check for typos in filename or path
- Use forward slashes (/) not backslashes (\) in paths
- Ensure no special characters in filenames

Image Not Found

Error: `! Package pdftex.def Error: File 'img/figure.png' not found`
Solution:

- Check that image file exists in specified path
- Verify file extension is correct (.pdf, .png, .jpg)
- Use relative paths from main .tex file location
- Ensure image is in supported format

Managing Large Documents

Compilation Time

For large theses with many figures:

- Use `\includeonly{chap/chapter_name}` to compile only specific chapters during editing
- Keep this in main .tex file, comment out when not needed
- Final compilation should include all chapters

Example:

```
% Uncomment to compile only specific chapters:
% \includeonly{chap/methodology,chap/results}
```

Draft Mode

For faster compilation during editing:

```
\documentclass[draft]{scrreprt}
```

Draft mode:

- Shows boxes instead of images
- Marks overfull boxes clearly
- Compiles faster
- Remember to remove `draft` option for final version

Quality Checks

Before Final Submission

Run these checks:

1. **Complete compilation**: Full pdfLaTeX→ bibtex → makeglossaries → pdfLaTeX× 2
2. **No warnings**: Address all LaTeXwarnings in log
3. **References check**: All citations appear in bibliography
4. **Cross-references**: No "??" in document
5. **Figures**: All figures appear correctly, not missing
6. **Page numbers**: TOC page numbers match actual pages
7. **Spelling**: Run spell checker
8. **Formatting**: Consistent throughout document

PDF Quality

Verify final PDF:

- **File size**: Reasonable (< 50 MB typically)
- **Hyperlinks**: Internal links work correctly
- **Bookmarks**: Chapter bookmarks appear in PDF viewer
- **Fonts**: All fonts embedded properly
- **Images**: Clear and readable at 100% zoom

Best wishes for your writing,
Krishna Acharya,

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